

When Does a Classifier Help an LLM?

Classifier-Guided Prompting and Hybrid Classifier–LLM Models for Credit-Default Prediction

Rishi Datta
Amador Valley High School
Pleasanton, California, USA
rishidatta20@gmail.com

Anandaswarup Vadapalli
Research Spark Hub Inc.
Dublin, California, USA
anandaswarup.vadapalli@gmail.com

Abstract—Large Language Models (LLMs) are increasingly proposed for credit-risk assessment, but how best to combine an LLM with a conventional fitted classifier on tabular data remains unclear. We study this question on the *Default of Credit Card Clients* dataset using a fixed, class-imbalanced evaluation set of 2,000 held-out clients, with all comparisons reported using imbalance-aware metrics (precision, recall, F1, AUC-ROC, AUC-PR) and bootstrap confidence intervals. Three findings emerge. First, a plain few-shot LLM (claude-sonnet-5) is *recall-competitive*: it attains the highest recall (0.47) and F1 (0.50) of any single model, exceeding a fitted Random Forest (recall 0.37, F1 0.48) and logistic regression (recall 0.25), but it is inferior at *ranking* (AUC-ROC 0.72 vs. 0.79). Second, instructing the LLM to *imitate* a classifier—by role (“act as a Random Forest”), by listing important features, or both—yields *no statistically significant* improvement over direct prompting; in contrast, using the classifier to construct the prompt does: pruning the prompt to the classifier’s most important features significantly improves recall (+0.071) and F1 (+0.032), while curating the few-shot examples (by decision-boundary proximity or by prototypicality) significantly *hurts*. Third, and most useful in practice, feeding the classifier’s predicted probability into the prompt (a hybrid model) raises the LLM’s AUC-ROC from 0.72 to 0.78—matching the Random Forest—while retaining significantly higher recall (+0.118 over the Random Forest). The reverse direction (LLM output as a feature for the classifier) yields no gain, and adding more classifier opinions (logistic regression and AdaBoost) does not improve on using the Random Forest alone. We additionally find that diffusion-based minority-class augmentation improves AUC-PR and precision over SMOTE but does not surpass simple class-weighting. Together these results give a concrete recipe: prune the prompt to a classifier’s important features and feed a single strong classifier’s probability into the LLM; do not rely on imitation instructions, example curation, or the reverse composition.

I. INTRODUCTION

Credit-risk assessment is a canonical supervised classification task: given a client’s repayment history, credit limit, billing and payment amounts, and demographics, a model estimates the probability of default. Logistic regression and tree ensembles such as Random Forests are long-standing, strong baselines because they capture the nonlinear interactions typical of credit data [1]–[3].

Large Language Models (LLMs) have attracted interest for financial decision support because they follow structured instructions and produce natural-language explanations [4]. However, an LLM presented with tabular records differs fundamentally from a *fitted* classifier: a trained classifier learns parameters from labeled data, whereas an LLM can only be *instructed* at inference time. Prior work has noted the difficulty of applying LLMs directly to numerical/tabular prediction and has generally kept prediction grounded in a conventional model while using the LLM for explanation [5]–[7].

This raises a practical question that, to our knowledge, has not been carefully isolated for credit risk: *when, and how, does a fitted classifier actually help an LLM (and vice versa)?* We distinguish three ways a classifier’s knowledge can be transferred to an LLM prompt and test each in a controlled manner:

- 1) **Imitation instructions:** tell the LLM to reason like the classifier (role and/or feature-importance hints).
- 2) **Classifier-guided prompt construction:** use the classifier to decide *what to put in the prompt*—which features to show, and which few-shot examples to use.
- 3) **Hybrid composition:** feed the classifier’s output (predicted probability and decision) into the prompt as input, or conversely feed the LLM’s output into the classifier.

A. Contributions

- A controlled comparison, on a fixed imbalanced evaluation set with imbalance-aware metrics and bootstrap confidence intervals, of a few-shot LLM against fitted logistic regression, Random Forest, and AdaBoost for credit-default prediction.
- An ablation showing that *imitation* prompting (classifier role, feature-importance hints, or both) gives no significant improvement over direct prompting.
- Evidence that *classifier-guided prompt construction* has opposite-signed effects: feature pruning significantly helps, while few-shot example curation (boundary or prototypical) significantly hurts.

- A hybrid model that feeds a classifier’s probability into the LLM, achieving Random-Forest-level ranking with significantly higher recall; and negative results for the reverse composition and for using multiple classifiers.
- A secondary study showing diffusion-based minority augmentation beats SMOTE on AUC-PR/precision but not simple class-weighting.

All comparisons that motivate a claim are accompanied by 95% bootstrap confidence intervals; we explicitly report null and negative results.

II. DATASET

We use the *Default of Credit Card Clients* dataset [1]: 30,000 clients, 23 features (credit limit, sex, education, marital status, age; repayment-status codes `PAY_0–PAY_6`; bill amounts `BILL_AMT1–6`; payment amounts `PAY_AMT1–6`), and a binary default label. The data are imbalanced: 22.1% default.

We split 80/20 with a fixed seed, *stratified* by label (24,000 train / 6,000 test). Fitted models train on the full 24,000. Because LLM inference is comparatively expensive, all LLM conditions are evaluated on a fixed, stratified subset of 2,000 held-out test clients (default rate 22.1%); every LLM condition sees the same 2,000 clients. The diffusion-augmentation study (Section IV-D), which involves only fitted models, is evaluated on the full 6,000-client test set.

III. METHODOLOGY

A. Fitted baselines

Majority predicts the majority (no-default) class. **Logistic Regression** is trained on standardized features. **Random Forest** (RF) uses 100 trees, maximum depth 12. **AdaBoost** uses 200 estimators. All use a fixed seed.

B. LLM prediction

The LLM is `claude-sonnet-5` with extended thinking disabled for near-determinism. Each client’s features are serialized as text; the model is given $k=12$ balanced labeled few-shot examples (6 default, 6 non-default) drawn from the training set and asked to return, for each client, a JSON object with a binary `label` and a `confidence` $\in [0, 1]$ interpreted as $P(\text{default})$ (used for AUC). Requests are batched (20 clients per call) and cached. We evaluate the following LLM conditions.

Imitation prompting.

- **L1 (direct)**: predict from features and the few-shot examples, no classifier guidance.
- **L2 (full)**: “act as a trained Random Forest” and a list of the RF’s top features to weigh.
- **L2a (role only)**: “act as a trained Random Forest,” no feature list.
- **L2b (features only)**: weigh the RF’s important features, no role.

Classifier-guided prompt construction (instruction text held constant at the plain L1 wording; only prompt *content* changes).

- **E1 (top- K features)**: include only the RF’s top-8 features; drop the other 15.
- **E2 (boundary shots)**: choose the 12 few-shot examples nearest the RF decision boundary (out-of-bag probability closest to 0.5), balanced by class.
- **E3**: E1 + boundary shots.
- **E4 (representative shots)**: choose the most confidently-correct examples (out-of-bag probability near 1 for defaulters, near 0 for non-defaulters), balanced.
- **E5**: E1 + representative shots.

Few-shot verdicts used for selection come from out-of-bag/cross-validated probabilities so they are not overfit.

Hybrid composition.

- **E6 (RF $\rightarrow$ LLM)**: E1 + the RF’s predicted probability and 0/1 decision as extra inputs for each client (and each few-shot example, using out-of-bag verdicts).
- **E7 (ensemble $\rightarrow$ LLM)**: E1 + the probabilities and decisions of *three* classifiers (RF, logistic regression, AdaBoost).
- **Inverse (LLM $\rightarrow$ RF)**: add the LLM’s E1 label and confidence as features to the RF; evaluated by 5-fold cross-validation on the 2,000 (comparing RF with vs. without the LLM feature on identical folds).

C. Metrics and uncertainty

We report accuracy, and—because the data are imbalanced—precision, recall, and F1 on the default class, plus AUC-ROC and AUC-PR. Overall accuracy is reported alongside the majority-baseline accuracy (0.779) to contextualize it. Uncertainty on the 2,000-client set uses 1,000 bootstrap resamples; comparisons between conditions use *paired* bootstrap (same resampled indices), and we call a difference significant when its 95% CI excludes zero. The diffusion study reports mean $\pm$ 95% CI across 5 seeds.

IV. RESULTS

Figure 1 previews the central trade-off: fitted models occupy the high-AUC/low-recall region, LLM conditions the low-AUC/high-recall region, and the hybrid E6 reaches the desirable high-AUC/high-recall corner.

A. LLM vs. fitted models, and imitation prompting

Table I reports all fitted baselines and the imitation-prompting LLM conditions on the 2,000-client set.

Two observations. (i) The LLM *trades precision for recall*: the direct LLM (L1) flags $\sim 20\%$ of clients as defaulters vs. the RF’s 12%, giving the highest recall (0.47) and, with E1, the highest F1 (0.53) of any model—above the fitted RF (0.48)—while the fitted models retain higher precision, accuracy, and AUC. The LLM is thus recall-competitive but ranking-inferior. (ii) **Imitation prompting does not help**: paired bootstrap differences of L2, L2a, and L2b versus L1 are non-significant on F1, AUC-ROC, and recall (all 95% CIs include zero; e.g., $L2-L1$ F1 = +0.004, CI $[-0.013, +0.020]$). Neither the “act as a Random Forest” role nor an explicit feature-importance list changes performance.

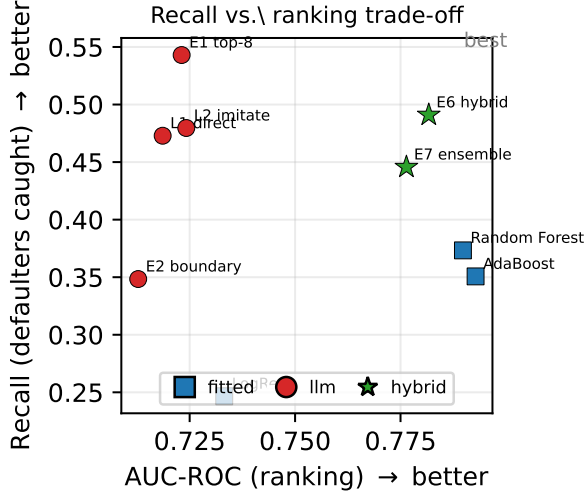


Fig. 1. Recall vs. AUC-ROC across models (N=2000). Fitted models (squares) rank well but catch few defaulters; plain/prompt-engineered LLM conditions (circles) catch more defaulters but rank worse; the hybrid E6 (star) attains Random-Forest-level ranking *and* high recall.

TABLE I

FITTED BASELINES AND IMITATION-PROMPTING LLM CONDITIONS (N=2000, DEFAULT RATE 22.1%). BEST VALUE PER COLUMN IN BOLD.

Model	Acc	Prec	Rec	F1	AUROC	AUPRC
Majority	0.779	—	0.000	0.000	—	—
Logistic Reg.	0.811	0.708	0.247	0.366	0.733	0.512
Random Forest	0.821	0.668	0.373	0.479	0.790	0.577
AdaBoost	0.823	0.698	0.351	0.467	0.793	0.571
L1: direct LLM	0.789	0.525	0.473	0.498	0.719	0.481
L2: role+features	0.790	0.526	0.480	0.502	0.724	0.478
L2a: role only	0.795	0.540	0.477	0.507	0.722	0.487
L2b: features only	0.790	0.526	0.489	0.506	0.728	0.486
E1: top-8 feats	0.786	0.515	0.543	0.529	0.723	0.470

TABLE II

PAIRED DIFFERENCES VS. L1 (MEAN [95% CI], 1000 BOOTSTRAP RESAMPLES). BOLD = SIGNIFICANT (CI EXCLUDES 0).

Condition	$\Delta F1$	$\Delta AUROC$	$\Delta Recall$
E1 top-8 feats	+0.032 [0.006, 0.056]	+0.005 [−0.013, 0.022]	+0.071 [0.040, 0.102]
E2 boundary shots	−0.057 [−0.088, −0.025]	−0.006 [−0.018, 0.006]	−0.125 [−0.158, −0.092]
E3 top-8+boundary	−0.011 [−0.039, 0.015]	+0.006 [−0.013, 0.023]	−0.045 [−0.075, −0.015]
E4 repr. shots	−0.029 [−0.054, −0.006]	+0.001 [−0.009, 0.013]	−0.070 [−0.097, −0.044]
E5 top-8+repr.	+0.020 [−0.005, 0.046]	+0.011 [−0.004, 0.027]	+0.016 [−0.012, 0.047]
E6 (E1+RF)	+0.026 [−0.000, 0.054]	+0.063 [0.039, 0.085]	+0.019 [−0.010, 0.048]
E7 (E1+RF+LR+ADA)	+0.026 [†]	+0.058 [†]	−0.010 [†]

[†]E7 differences vs. L1 shown for reference; E7 is compared against E6 in the text.

B. Classifier-guided prompt construction

Table II reports paired differences versus L1 for the prompt-construction and hybrid conditions.

Feature pruning helps; example curation hurts. Restricting the prompt to the RF’s top-8 features (E1) significantly improves recall (+0.071) and F1 (+0.032)—and E1 attains the best F1 of any model in the study (0.529), above the fitted

TABLE III

HYBRID AND REFERENCE MODELS (N=2000). E6/E7 vs. RF PAIRED: E6 RECALL +0.118 (SIG), F1 +0.045 (SIG), AUROC −0.008 (SIG). E7 vs. E6: RECALL −0.045 (SIG), F1/AUROC n.s.

Model	Acc	Prec	Rec	F1	AUROC	AUPRC
E1 (feats only)	0.786	0.515	0.543	0.529	0.723	0.470
E6 (E1+RF)	0.803	0.561	0.491	0.524	0.782	0.549
E7 (E1+RF+LR+ADA)	0.811	0.595	0.446	0.510	0.776	0.536
Random Forest	0.821	0.668	0.373	0.479	0.790	0.577

RF. In contrast, both example-curation strategies significantly *reduce* recall and F1: boundary (hardest-case) selection (E2) is worst ($\Delta F1$ −0.057, $\Delta recall$ −0.125), and representative (prototypical) selection (E4) also significantly hurts ($\Delta F1$ −0.029). Combining feature pruning with curated examples (E3, E5) dilutes E1’s gain to non-significance. The takeaway is that a classifier is useful for deciding *which features* the LLM should see, but random balanced few-shot examples are better than RF-curated ones.

C. Hybrid classifier–LLM composition

RF → LLM (E6) is the best operating point. Adding the RF’s probability and decision to the prompt raises AUC-ROC from 0.719 (L1) to 0.782—a +0.063 significant gain that essentially matches the fitted RF (0.790)—and AUC-PR from 0.481 to 0.549. Crucially, E6 retains high recall: versus the fitted RF, E6 improves recall by +0.118 and F1 by +0.045 (both significant), at a negligible AUC cost (−0.008). Thus E6 attains Random-Forest-level ranking with substantially higher recall (Table III).

More classifiers do not help. Feeding three classifier opinions (RF, logistic regression, AdaBoost; E7) does not improve ranking over the RF-only hybrid (E7 vs. E6: $\Delta AUROC$ −0.005, n.s.; $\Delta F1$ −0.014, n.s.) and significantly *reduces* recall (−0.045). The three classifiers are highly correlated (all AUC-ROC $\approx$ 0.79, all high-precision/low-recall), so their combined signal is redundant and pushes the LLM toward a more conservative consensus.

The reverse composition does not help. Adding the LLM’s E1 output (label and confidence) as features to the RF, evaluated by 5-fold cross-validation on the 2,000 clients, yields no significant change: RF-with-LLM versus RF-without on identical folds gives $\Delta recall$ +0.002, $\Delta F1$ −0.004, $\Delta AUROC$ −0.002 (all n.s.). Notably the RF ranks the LLM’s confidence as its most important input, yet metrics do not move—the LLM’s signal, being derived from the same features, is redundant with what the RF already extracts. Information useful for the hybrid flows one way: the classifier supplies ranking ability the LLM lacks, but the LLM supplies nothing the classifier lacks.

D. Diffusion-based minority augmentation

Because recall is limited by class imbalance, we test whether augmenting the training data with *synthetic defaulters* from a diffusion model helps a fitted RF, compared with standard remedies. We train a Gaussian denoising diffusion

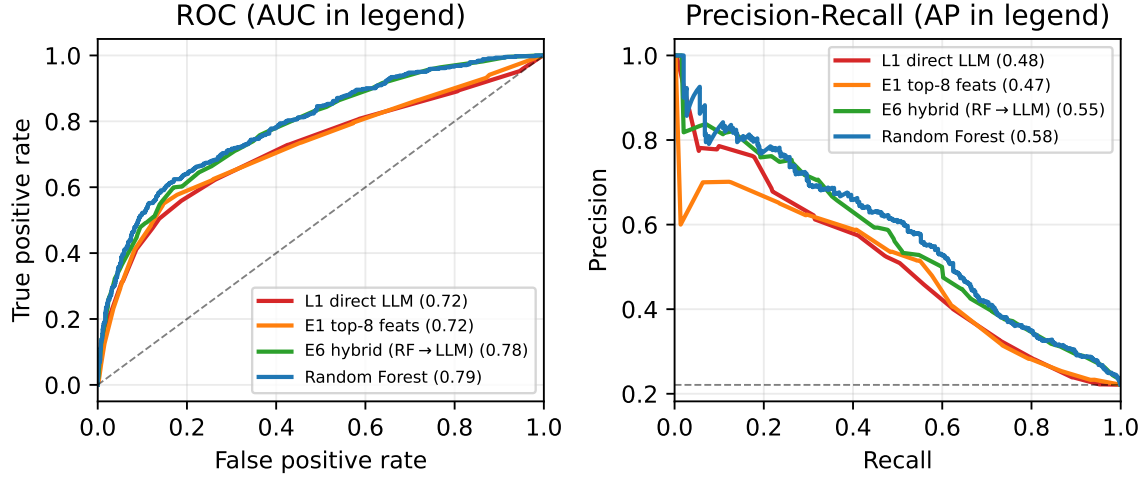


Fig. 2. ROC (left) and precision-recall (right) curves on the 2000-client test set for the direct LLM (L1), feature-pruned LLM (E1), the hybrid (E6), and the fitted Random Forest. Feeding the RF’s probability into the prompt (E6) lifts the LLM’s ROC/PR curves to essentially the Random Forest’s level.

TABLE IV
RANDOM FOREST UNDER IMBALANCE REMEDIES (FULL TEST, N=6000;
MEAN $\pm$ 95% CI OVER 5 SEEDS).

Strategy	Recall	F1	AUPRC
original	0.351 $\pm$.004	0.458 $\pm$.003	0.556 $\pm$.003
class_weight	0.575 $\pm$.004	0.538 $\pm$.003	0.550 $\pm$.003
oversample	0.561 $\pm$.005	0.537 $\pm$.003	0.547 $\pm$.001
SMOTE	0.541 $\pm$.006	0.511 $\pm$.005	0.518 $\pm$.004
diffusion	0.541 $\pm$.045	0.526 $\pm$.017	0.540 $\pm$.015

model (with a quantile-normal transform of continuous features) on the defaulter rows and generate synthetic defaulters to rebalance the training set; a detection classifier distinguishes real from synthetic defaulters with AUC 0.80 (imperfect but usable). Table IV reports RF performance on the full 6,000-client test set, averaged over 5 seeds.

Paired across seeds, diffusion augmentation *significantly* beats SMOTE—the standard synthetic-minority baseline—on AUC-PR (+0.022), precision (+0.028), and accuracy (+0.013), while tying on recall and F1. However, it does *not* beat simple class-weighting, which is free and gives the highest recall/F1, and diffusion shows higher run-to-run variance. Diffusion augmentation is therefore a better synthetic generator than SMOTE for this task but is not worth its complexity relative to class-weighting.

V. DISCUSSION

Our results give a consistent picture of how a fitted classifier and an LLM should—and should not—be combined for imbalanced tabular credit prediction.

Telling an LLM to imitate a classifier is ineffective, but *using* the classifier to shape the prompt is effective in one specific way: selecting which features to show. This is intuitive—dropping 15 low-importance columns reduces distraction—and it is the only prompt-only manipulation that significantly

helped. Curating few-shot examples, by either difficulty or prototypicality, hurt; random balanced examples were consistently better, suggesting that example *diversity* matters more than example “informativeness” for this task.

The strongest practical result is the hybrid: feeding a single strong classifier’s probability into the prompt gives the LLM the ranking ability it otherwise lacks (AUC-ROC 0.72 $\rightarrow$ 0.78) while preserving its higher recall—yielding a model that ranks like the Random Forest but catches substantially more defaulters. Because Random Forest inference is essentially free, this hybrid is cheap to deploy. The asymmetry is informative: the reverse composition (LLM $\rightarrow$ classifier) and richer ensembles (three classifiers $\rightarrow$ LLM) added nothing, indicating that a single strong classifier already supplies the complementary signal the LLM needs.

VI. LIMITATIONS

Results use a single LLM (claude-sonnet-5), a single dataset, one 2,000-client evaluation draw, and one few-shot draw per condition. Bootstrap CIs capture test-set sampling uncertainty but not LLM-sampling or few-shot-draw variance; the ablation and hybrid conclusions would be strengthened by multiple prompt draws and additional models. The inverse-composition test trains the cross-validated RF on $\sim 1,600$ rows per fold, below the 24,000-row RF, though the RF-with vs. RF-without comparison is controlled. The LLM used only the same features available to the classifiers; an LLM with access to information the classifier lacks (e.g., unstructured text) might contribute complementary signal, which we did not test. The diffusion generator treats ordinal repayment codes as continuous; a fully mixed-type generator might improve synthetic quality.

VII. CONCLUSION

For imbalanced tabular credit-default prediction, a few-shot LLM is recall/F1-competitive with fitted models but inferior

at ranking. Instructing the LLM to imitate a classifier does not help; the effective recipe is to (i) prune the prompt to the classifier’s most important features and (ii) feed a single strong classifier’s predicted probability into the prompt. The resulting hybrid matches the Random Forest’s ranking while significantly improving recall, at negligible added cost. Curating few-shot examples, adding more classifier inputs, and the reverse (LLM-into-classifier) composition all failed to help. Separately, diffusion-based minority augmentation outperforms SMOTE but not class-weighting. These findings favor cheap, classifier-grounded hybrids over LLM-imitation prompting for credit risk.

Diffusion generator: Gaussian DDPM (cosine schedule) on quantile-normal-transformed defaulter rows.

ACKNOWLEDGMENT

The authors thank Research Spark Hub Inc. for support in the preparation of this work.

REFERENCES

- [1] I.-C. Yeh and C.-h. Lien, “The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients,” *Expert Systems with Applications*, vol. 36, no. 2, pp. 2473–2480, 2009.
- [2] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [3] S. Lessmann, B. Baesens, H.-V. Seow, and L. C. Thomas, “Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research,” *European Journal of Operational Research*, vol. 247, no. 1, pp. 124–136, 2015.
- [4] T. B. Brown *et al.*, “Language models are few-shot learners,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 1877–1901, 2020.
- [5] M. Golec *et al.*, “Interpretable and explainable machine learning for credit risk: challenges with large language models,” *arXiv preprint*, 2025.
- [6] M. K. Nallakaruppan *et al.*, “Credit risk assessment with explainable artificial intelligence,” *IEEE Access*, 2024.
- [7] N. Bussmann, P. Giudici, D. Marinelli, and J. Papenbrock, “Explainable machine learning in credit risk management,” *Computational Economics*, vol. 57, pp. 203–216, 2021.

APPENDIX A

PROMPT AND REPRODUCIBILITY DETAILS

All clients were serialized as comma-separated name=value fields. The base prompt states the task, briefly defines the feature groups (PAY_* repayment codes, BILL_AMT, PAY_AMT, LIMIT_BAL), lists $k=12$ balanced labeled examples, gives the condition-specific instruction, lists the clients, and requires a JSON array of `{id,label,confidence}` with no prose. In E1/E3/E5 the serialization is restricted to the RF’s top-8 features (PAY_0, PAY_2, BILL_AMT1, PAY_4, PAY_3, LIMIT_BAL, PAY_AMT1, BILL_AMT2); in E6/E7 each row additionally carries classifier probabilities and 0/1 decisions (out-of-bag/cross-validated for few-shot examples, fitted-model outputs for test clients). Data split: stratified 80/20, seed 42; LLM evaluation on a stratified 2,000-client subset of the test set. Fitted models: RF (100 trees, depth 12), logistic regression (standardized), AdaBoost (200 estimators). LLM: `claude-sonnet-5`, extended thinking disabled, batched at 20 clients per request with response caching. Uncertainty: 1,000 bootstrap resamples (paired for between-condition comparisons); diffusion study over 5 seeds.